

Experimental Identification of Diffusive Coupling Using 2D NMR

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Spin relaxation based nuclear magnetic resonance (NMR) methods have been used extensively to determine pore size distributions in a variety of materials. This approach is based on the assumption that each pore is in the fast diffusion limit but that diffusion between pores can be neglected. However, in complex materials these assumptions may be violated and the relaxation time distribution is not easily interpreted. We present a 2D NMR technique and an associated data analysis that allow us to work directly with the time dependent experimental data without Laplace inversion to identify the signature of diffusive coupling between different pores. Measurements on microporous glass beads and numerical simulations are used to illustrate the technique.

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NMR has been used to study a wide array of biological, geological, and synthetic porous materials [1,2]. In these systems the decay of the spin magnetization is controlled by a combination of diffusion and enhanced relaxation at the pore-solid interface. If the surface relaxation is relatively weak, then each pore will exhibit single exponential decay with a lifetime determined by its size; this is the fast diffusion regime, and the net decay signal is a sum of exponentials that reflects the distribution of pore sizes. However, these assumptions can break down, either because the surface relaxation is large enough that relaxation within a given pore involves several modes or because inter-pore diffusion is important. For example, in materials with a mixture of small and large pores, the 1D relaxation time spectrum can be either very narrow or very broad, depending on the connectivity between pores of different sizes [3]. The situation is further complicated by the presence of spatial heterogeneity of pore sizes over distances beyond the diffusion length (typically $\sim 100 \mu\text{m}$) [4]. This Letter presents a 2D NMR relaxation method to identify the diffusion regimes.

Two-dimensional NMR experiments involve a correlated measurement of longitudinal (T_1) and transverse (T_2) relaxation and a 2D correlation map (e.g., T_1 - T_2) is obtained via a 2D Laplace inversion [5–9]. In these experiments diffusive coupling and pore size distributions can exhibit qualitatively different features. For example, when all of the pores are in the fast diffusion regime, the signal from each pore will exhibit a single T_1 and T_2 value and thus appears as a single point on the 2D correlation plot. A distribution of pore sizes will lead to signals along a ridge in the 2D plot with the specific ratio $r \equiv T_1/T_2$. On the other hand, if the fast diffusion regime does not apply, the 2D T_1 - T_2 cross plot can be considerably more complicated with significant contributions away from the $T_2 = rT_1$ ridge. Indeed, it has been shown theoretically that in certain regions the signal can have small negative

amplitude peaks, a fact that severely limits the stability of the 2D Laplace inversion [7,10,11]. Therefore, a new approach to the T_1 - T_2 measurement is needed for complex systems. We introduce a way of working directly with the time-domain data that can sense the small negative signals that are associated with more complex diffusion phenomena. This technique is illustrated with a study of packed microporous glass beads where diffusion occurs between the intergranular pores and the micropores inside the grains. Our measurements and numerical simulations of the diffusion dynamics are shown to identify the presence of the negative correlation peaks. Our experiments were exclusively performed at low magnetic field (~ 500 G) to avoid the complication of higher magnetic fields [12].

The evolution of the nonequilibrium magnetization, $m(\mathbf{r}, t)$, is described by the diffusion equation [13],

$$D_0 \nabla^2 m(\mathbf{r}, t) = \frac{\partial m(\mathbf{r}, t)}{\partial t}, \quad (1)$$

subject to the boundary condition on the interface between the pore fluid and the solid grains,

$$[D_0 \hat{\mathbf{n}} \cdot \nabla m(\mathbf{r}, t) + \rho m(\mathbf{r}, t)]_S = 0. \quad (2)$$

Here D_0 is the intrinsic diffusion coefficient for the fluid saturating the pores, and the parameter ρ measures the strength of the enhanced relaxation at the interface, S . For a pore of size a , a key dimensionless parameter is $k \equiv \rho a / D_0$ [14]. In the fast diffusion regime, where the fluid molecules diffuse extensively before being relaxed at the interface, we have $k \ll 1$ and the magnetization decays as a single exponential with lifetime $T \sim a / \rho$. For $k \geq 1$ (i.e., larger pores or stronger surface relaxation) spin relaxation becomes a multiexponential process.

An example in Fig. 1 shows the T_2 spectra of two samples of randomly packed solid borosilicate glass beads of diameter 55 and 150 μm , respectively. From the measured

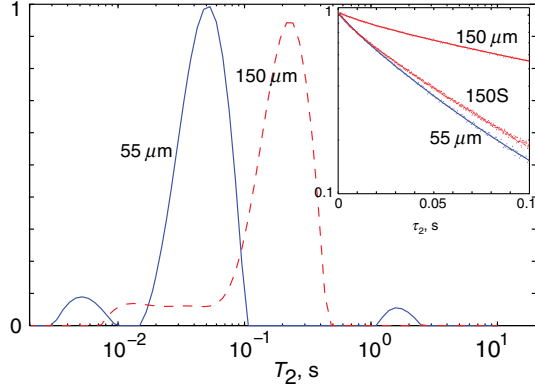


FIG. 1 (color online). T_2 spectrum for two packings of solid borosilicate beads with diameters of 55 and 150 μm . The inset shows the initial echo decay data for the two samples. The trace marked 150S is a scaled decay of the 150 μm sample to match the initial decay rate of the 55 μm sample demonstrating the higher curvature of the decay of the large-grain sample.

relaxation we estimate $k = 1.7$ and 4.4 for the two samples. Thus the 150 μm system is certainly not expected to exhibit fast diffusion behavior, while the 55 μm sample is in an intermediate regime. This is confirmed by the measured T_2 distributions; the 150 μm sample shows a more pronounced tail at short T_2 than does the 55 μm sample. Such behavior can also be seen in the time-domain decay (Fig. 1 inset) where the larger-grain data show more curvature. We emphasize that the short- T_2 tail does not directly represent the small pores; instead, it reflects the complicated diffusion dynamics in the pore space [14]. Even though this forward problem is well understood, the inverse problem, on the other hand, is much more difficult. For example, given a broad T_2 distribution from a natural material, it is impossible to know from this 1D experiment whether slow diffusion is the source of the distribution's width or if the width simply reflects a wide distribution of pore sizes.

This Letter focuses on the T_1 - T_2 correlation experiment which uses the inversion-recovery-Carr-Purcell-Meiboom-Gill pulse sequence [8],

$$\text{RD} - \pi - \tau_1 - \frac{\pi}{2} - t_e/2 - [\pi - t_e/2 - \text{Acq} - t_e/2]_N, \quad (3)$$

where N is the echo number, RD the recycle delay, and Acq represents data acquisition. The signal is obtained in a 2D matrix $\mathbf{m}(\tau_1, \tau_2 \equiv Nt_e)$. We define $\mathbf{M}(\tau_1, \tau_2) \equiv \mathbf{m}(\tau_1 \rightarrow \infty, \tau_2) - \mathbf{m}(\tau_1, \tau_2)$, which is the deviation of signal from its equilibrium. $\mathbf{M}$ is related to the T_1 - T_2 correlation function $\mathbf{S}(T_1, T_2)$ by

$$\mathbf{M}(\tau_1, \tau_2) = \int dT_1 dT_2 \mathbf{S}(T_1, T_2) \exp(-\tau_1/T_1 - \tau_2/T_2) \quad (4)$$

The formalism used by Brownstein and Tarr has been extended to characterize 2D experiments [7,10]. The magnetization solution is written as

$$m(\mathbf{r}, t) = \sum_{n=0}^{\infty} a_n \phi_n(\mathbf{r}) e^{-t/T_n}, \quad (5)$$

and the detected signal is the volume integral of m . Here ϕ_n and T_n are the eigenfunctions and eigenvalues of Eq. (1) and the boundary condition [Eq. (2)]. It is important to note that the surface relaxation parameters governing longitudinal and transverse relaxation, ρ_1 and ρ_2 ($\rho_1 < \rho_2$), are different [15]. Thus two distinct eigensystems are required to describe the T_1 - T_2 experiment. We denote these as $\{\phi_{1,n}, T_{1,n}\}$ and $\{\phi_{2,n}, T_{2,n}\}$. The difference between the two eigensystems is critical to the phenomena discussed in this Letter. The T_1 - T_2 correlation spectrum can be written as [7]

$$\begin{aligned} \mathbf{S}(T_1, T_2) = & \sum_{p,q} \langle f | \phi_{2,q} \rangle \langle \phi_{2,q} | \phi_{1,p} \rangle \langle \phi_{1,p} | i \rangle \\ & \times \delta(T_1 - T_{1,p}) \delta(T_2 - T_{2,q}), \end{aligned} \quad (6)$$

where $\langle \phi | \psi \rangle \equiv \int d^3\mathbf{r} \phi^*(\mathbf{r}) \psi(\mathbf{r})$ and $|i\rangle$ and $|f\rangle$ represent the initial and final distributions which are taken to be constant. While $\langle f | \phi \rangle$ and $\langle \phi | i \rangle$ are associated with the initial and final states, the cross terms $\langle \phi_{2,q} | \phi_{1,p} \rangle$ are related to the overlap of the eigenfunctions in the transverse and longitudinal manifolds and are not detected in 1D experiments [14]. For each value of T_1 and of T_2 , there may be a nonzero value of $\mathbf{S}(T_1, T_2)$, even if $T_1 < T_2$. This simply reflects the ability of a longitudinal mode with relaxation time T_1 to couple into a transverse mode with relaxation time T_2 . For example, terms in Eq. (6) with $p \neq q$ are referred to as off diagonal because they contribute to $\mathbf{S}(T_1, T_2)$ away from the $T_1 = T_2$ diagonal line, which is dominated by terms with $p = q = 0, 1, 2, \dots$. Figure 2 shows an example from Ref. [7], a single, one-dimensional pore with $k_1 \equiv \rho_1 a/D = 0.5$ and $k_2 \equiv \rho_2 a/D = 5$. In this case the eigenmodes were evaluated analytically and $\mathbf{S}(T_1, T_2)$ was obtained from Eq. (6), without Laplace inversion. In addition to contributions near the diagonal line, Fig. 2(a) clearly shows off-diagonal terms corresponding to $p = 0$ and $q = 1$, etc. Note that the amplitude of the off-diagonal terms is small and that the peaks above the diagonal are negative; thus, the conventional algorithm [8] cannot apply.

Small off-diagonal signals are difficult to detect because of the ill-conditioned nature of Laplace inversion techniques [8]. However, the negative character of such peaks has a signature that can be seen directly in the time domain signal without using Laplace inversion, as shown by the dashed line in Fig. 2(b). Note that, for a larger τ_2 , $\mathbf{M}$ initially increases as τ_1 increases and then decays. The slope of $\mathbf{M}$ at small τ_1 is $-\int dT_1 \mathbf{S}(T_1, T_2)/T_1$. Since $T_1 > 0$, a positive initial slope (as observed in Fig. 2) indicates that the amplitude $\mathbf{S}(T_1, T_2)$ contains negative peaks which can only arise in systems that are not in the fast diffusion regime.

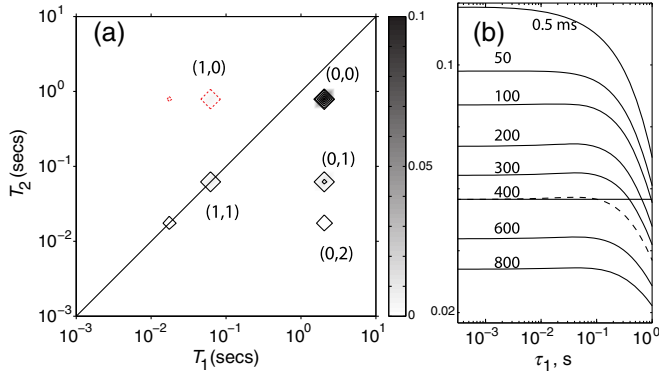


FIG. 2 (color online). T_1 - T_2 data from Fig. 1 of Ref. [7]. (a) T_1 - T_2 spectrum showing several peaks $[(p, q), \text{marked}]$ and negative signals above the diagonal (dashed contour lines). (b) The time domain signal M is plotted as a function of τ_1 at several τ_2 values (marked in the panel). Note the small initial rise of the signal at longer echoes highlighted by the dashed line.

T_1 - T_2 measurements were carried out on a water saturated pack (1 in. diameter and 1 in. long) of about $150 \mu\text{m}$ microporous beads (CPG3000A, Millipore), and on a bulk water sample for comparison (Fig. 3). The CPG beads are themselves porous, with an average micropore diameter of $0.3 \mu\text{m}$. For this sample the total porosity is 80% and is divided equally between the micro- and macropores. The NMR experiments were performed using a Rock Core Analyzer (Magritek Limited, Wellington, NZ) at ^1H NMR frequency of 2 MHz. Because of the large surface area, spin relaxation for the water inside the grains is substantially enhanced. The water molecules in the macropores (between the grains) are relaxed by diffusion into the micropores. Because diffusion between the micropores and the macropores is an essential part of the overall relaxation process, this system cannot be described with a picture based on fast diffusion.

In Figs. 3(b) and 3(c) the decay data, $M(\tau_1, \tau_2)/M(0, \tau_2)$, is graphed as a function of τ_1 for the different echo times, and each trace is shifted vertically in order to highlight the details at short τ_1 . For the bulk water sample [panel (c)], we see a continuously decreasing signal for all echo times. By contrast, in Fig. 3(b), the signal is observed to increase for $\tau_2 > 240$ ms. As discussed above, this behavior is a clear indication of negative peaks in the T_1 - T_2 spectrum. Furthermore, since the positive slope in M occurs when τ_1 is small, the negative amplitude in $S(T_1, T_2)$ must be at low T_1 values. Similarly, since the positive slope is seen only at longer echo times, we expect the negative amplitudes to be associated with large T_2 values (≥ 240 ms). Thus, looking only at the time-domain data, without any Laplace inversion, we can argue that the negative contributions have to be in the upper left corner of the T_1 - T_2 spectrum, consistent with the analytic result of Fig. 2.

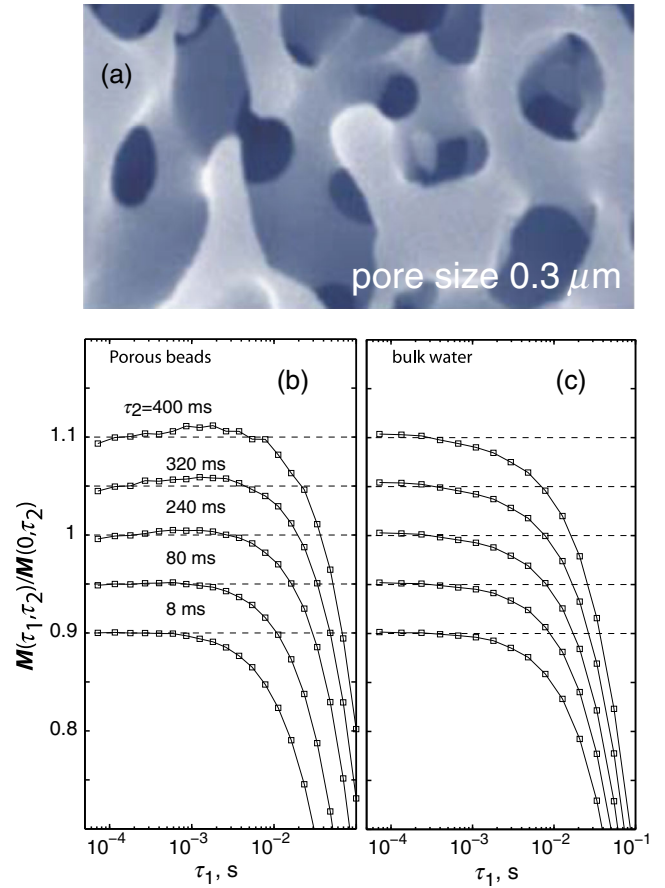


FIG. 3 (color online). (a) A micrograph shows the internal structure of the porous beads. The light gray is the solid phase. Average pore diameter is $0.3 \mu\text{m}$. (b), (c) T_1 - T_2 measurements on porous beads (b) and water (c). $M(\tau_1, \tau_2)$ is plotted as a function of τ_1 for the different echo times shown above each data set. The data are scaled to the average of the first eight data points and then shifted away from unity for display. An initial rise is observed for echo times larger than 240 ms. This rise corresponds to negative components in $S(T_1, T_2)$ associated with diffusive coupling between the micro- and macropores. No initial rise is observed for the bulk water signal.

Relaxation typically describes signal loss as time increases; thus, it is surprising to observe signal increase as τ_1 lengthens. To understand the physical origin of the observed phenomenon, consider the highly simplified model shown in Fig. 4, with a microporous region (shaded) to the left of the macropore. The qualitative behavior of the T_1 modes [4(a)] is central to this phenomenon. The slowest mode, $\phi_{1,0}$, which would be spatially uniform in the limit $\rho_1 \rightarrow 0$, has the same sign in both regions. Because the source of relaxation is the micropores, the amplitude of $\phi_{1,0}$ is higher in the macropore and lower in the micropore. The second mode, $\phi_{1,1}$, is asymmetric and positive in the micropore and negative in the macropore due to its orthogonality with the first mode. Figure 4(a) shows the contribution to the initial magnetization (m_0) from $\phi_{1,0}$ (the long dashed line) and $\phi_{1,1}$ (the short dashed line), and they

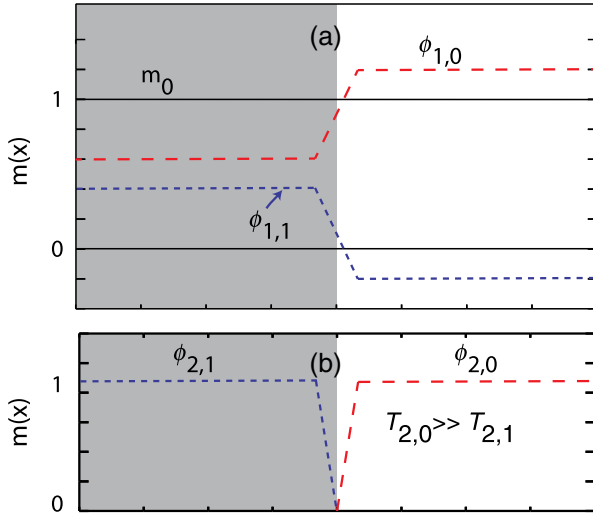


FIG. 4 (color online). Qualitative illustration of the first two eigenmodes of T_1 (a) and T_2 (b) in macro- and micropores. The T_2 modes are mostly isolated to macro- and micropores due to the larger ρ_2 .

add up to the constant m_0 , $m_0 = a_0\phi_{1,0} + a_1\phi_{1,1}$. Note that $|a_0\phi_{1,0}| > m_0$ in the macropore region [Fig. 4(a)].

We further assume ρ_2 is much larger than ρ_1 , and thus the T_2 modes behave as if the two pores are mostly decoupled [Fig. 4(b)]. As a result, at long τ_2 the micropore signal decays and the detected signal only comes from the macropore, thus

$$m \propto e^{-(\tau_1/T_{1,0})}|a_0\phi_{1,0}| - e^{-(\tau_1/T_{1,1})}|a_1\phi_{1,1}|, \quad (7)$$

evaluated within the macropore. Note that $T_{1,0} > T_{1,1}$. As τ_1 increases, the weight of $|\phi_{1,1}|$ decreases and thus $\mathbf{M}$ increases. For our sample, the contrast between ρ_1 and ρ_2 is finite; thus, the resulting increase is small.

To further validate our argument, we have performed numerical simulations based on a three-dimensional microporous model whose geometrical parameters can be matched to those of the bead pack discussed in connection with Fig. 3. The simulations employ variable step-size random walks in a packing of microporous spherical grains [16,17]. The large grains are arranged on a face centered cubic lattice; by adjusting their radius we can fix the amount of macroporosity at 0.40. We have examined two ways of representing the microporous structure within these grains based on periodic arrangements of either spherical micrograins or overlapping spherical micropores. In both cases the porosity within the large grains is adjusted to 0.67 so that the porosity contribution from the macro- and micropores are both 0.40. The grain sizes are adjusted so that the effective pore volume to surface area ratio, $V_P/S = 0.31 \mu\text{m}$. The diffusion coefficient for the bulk fluid is $2.5 \mu\text{m}^2/\text{ms}$. First, the comparison of the measured diffusion coefficient $[D(t)]$ with the simulation [16],

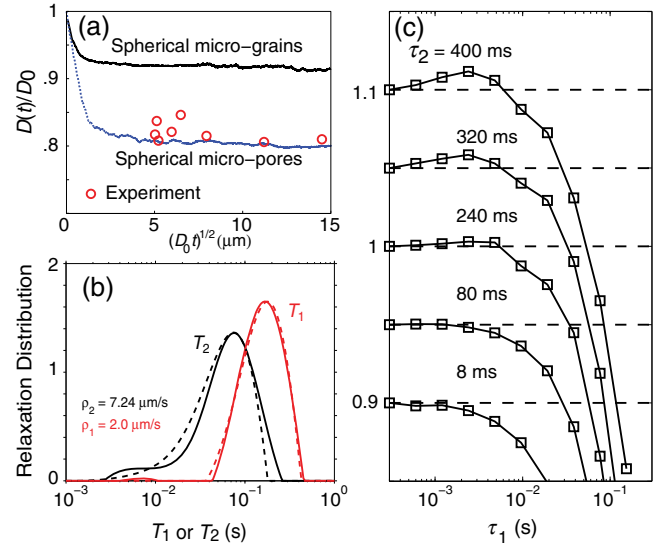


FIG. 5 (color online). Simulation results for microporous bead models. (a) Comparison of the measured time dependent diffusion coefficient, $D(t)$, with calculations based on two models for the microstructure. (b) Comparison of the simulated T_1 and T_2 distributions (dashed lines) with the measurements (solid lines). (c) Simulation of the T_1 - T_2 experiment. Data are plotted as in Fig. 3(b), showing the initial rise for $\tau_2 > 240$ ms.

Fig. 5(a), shows that the model with spherical micropores provides a much better fit to the experimental diffusion data. Second, the simulation of the T_1 and T_2 distributions is compared with our experiment [Fig. 5(b)] in order to determine ρ_1 and ρ_2 . Next, the micropore model was used to simulate the T_1 - T_2 experiment and the results are presented in Fig. 5(c). The calculations are presented in the same fashion as the experimental data [Fig. 3(b)] and a small positive slope is observed for $\tau_2 > 240$ ms, as in the experiment.

Consistency between our measurements, simulation, and analysis demonstrates that the T_1 - T_2 experiment can identify deviations from the fast diffusion regime. Evidence for this effect can be seen directly in the time-domain data without the complication of the Laplace inversion. In the presence of a broad spectrum of pore sizes, this method should continue to work well because $\mathbf{S}(T_1, T_2)$ integrates contributions from different pore sizes and the negative contributions will remain. The position of the negative structure is likely to depend on the pore size and will produce a strip of off-diagonal negative amplitude (instead of individual peaks). The location and magnitude of this structure will hold the information about pore coupling in such systems.

In conclusion, we have demonstrated that time dependent 2D NMR data can be used directly to identify complex diffusion regimes and diffusive coupling between pores in porous media. We plan to extend our analysis to other 2D experiments (e.g., T_2 -storage- T_2 experiments) [7,17] and natural samples with a wide distribution of pore sizes.

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